

Lab 2

Course goals

- To implement and use data structures and algorithms in application programs.
 - To implement a dynamic data structure, doubly linked list, using concepts such as C++ classes with overloaded operators and dynamic memory allocation
- To analyze and evaluate various data structures in solving computational problems with respect to efficiency and appropriateness

Preparation

Please ensure you read the entire lab description thoroughly, before starting any exercises, including the preparation steps given below. Pay special attention to the [code requirements section](#).

Completing the following tasks before the Lab2 HA session will help you make the most of your lab time

- Download the [files for this exercise](#) from the course website. Like lab 1, you can then use CMake to create a project for this lab.
- Compile, link, and execute the program. The first assertion should fail.
- Review lectures 2 to 4. Big-O notation was introduced in [lecture 2](#) and [lecture 3](#), while doubly linked lists were discussed in [lecture 4](#).
- Study the class Set interface given in the file `set.h`.
- Some member functions of class Set are explicitly marked as “// IMPLEMENT before Lab2 HA”, see files `set.h` and `set.cpp`. At least, these functions should be implemented before the HA lab session.

You can test your code with the program given in `lab2.cpp`. Your code should pass the tests in phase 0 to phase 6, before HA lab session. The expected output is shown [here](#).

- Understand the algorithm for [merging two sorted sequences](#) given in the appendix. This algorithm **should be** used to implement the union of two sets (i.e. member function `Set::operator+=`) in linear time. This algorithm was also used in lab1 of [TNG033 course](#).

If you have any specific question about the exercises, then send us an e-mail. Be short and concrete, otherwise you won't get a quick answer. You can write your e-mail in Swedish. Add the course code to the e-mail's subject, i.e. “TND004: ...”.

A class to represent sets using doubly linked lists

In the [TNG033 course](#), there was a lab about implementing a singly linked list to represent sets of integers. In this lab exercise, you go further and implement a class Set using instead **sorted doubly linked lists**. Additionally, it is required that all Set functions have **linear time complexity, in the worst-case**. Therefore, you should

be careful when re-using code from your lab solution in TNG033 for this exercise, since TNG033 did not require each Set operation to execute in linear time¹.

Course book

Section 3.5 of the course book presents a possible implementation for doubly linked lists. Though it is also required to implement doubly linked lists in this lab, there are some important differences between the implementation required here and the book's implementation, as summarized below. Most of the differences are motivated by the fact that we are going to use doubly linked lists to implement the concept of (mathematical) set. Nevertheless, studying the implementation presented in section 3.5 of the course book is recommended.

- The book presents an implementation of doubly linked lists, while in this lab you are requested to implement the concept of set by using a doubly linked list (though, sets can be implemented in other ways).
- In this lab, list nodes must be kept in sorted order and duplicates are not allowed. In contrast, the course book's implementation permits duplicates and does not require the list to be sorted.
- The book uses a template class to implement a generic class List of objects. Template classes are not used in this lab.
- Iterator classes are not considered in this exercise, though they are implemented in the course book.
- Move constructor and move assignment operator are part of the book's implementation (lines 26 to 41 of figure 3.16 of course's book). These are not part of the lab.
- The implementation of class Node provided with this lab maintains a counter of the total number of existing nodes (a static data member called `count_nodes`).
- This lab follows C++20 approach with respect to overloading comparison operators.

Counting nodes and assertions

A (static) member function named `get_count_nodes` of class Set returns the total number of existing nodes. This function is used in the test code to help detecting possible memory leaks by using assertions.

Similar to lab 1, assertions are used to help test your code. A test program is given in the main function (see `lab2.cpp`). For instance, the assertion

```
assert(Set::get_count_nodes() == 2);
```

tests whether the total number of existing nodes is equal to 2. If not then the program stops running and a message is displayed with information about the failed assertion (see [lab 1](#), section “Testing the code: assertions”).

¹ In other words, your code for the Set class in TNG033 course may not satisfy the linear time requirement of this lab and, consequently, cannot be simply re-used.

Neither the data member `count_nodes` nor function `get_count_nodes` should be used in the member functions implementations unless you want to add some extra tests.

Exercise 1: class Set

In this exercise, you need to implement the class `Set` that represents sets of integers (`int`). Sorted doubly linked list is the data structure used to implement sets in this lab. Notice that sets do not have repeated elements.

Every node of the list stores an integer value. To make it easier to remove and insert an element from the list, the list's implementation uses "dummy" nodes at the head and tail of the list, as discussed in [lecture 4](#) and in the course book. Thus, an empty doubly linked list consists of two nodes pointing at each other (see also figure 3.10 of course book).

The class `Set` provides the usual set operations like set union, difference, subset test, and so on. The class definition is given in the file `set.h`. Extra information about the meaning of the overloaded operators is given below.

- Overloaded operator `==` such that `R == S` returns `true`, if R and S represent the same set (i.e. both sets have the same elements). Otherwise, `false` is returned.
- Overload the three-way comparison operator `<=>`. See the appendix for an introduction to [C++20 comparison operators](#). Implementing this operator enables comparisons between two sets, such as testing for equality (`s1 == s2`) and subset relations (`s1 <= s2`)².
- Overloaded operator `+=` such that `R += S`; should be equivalent to $R = R \cup S$, i.e. the *union* of R and S is assigned to set R . Recall that the union $R \cup S$ is the set of elements in set R or in set S (without repeated elements). For instance, if $R = \{1, 3, 4\}$ and $S = \{1, 2, 4\}$ then $R \cup S = \{1, 2, 3, 4\}$.

Union of sets is conceptually like the problem of merging two sorted sequences. An algorithm to merge sorted sequences efficiently is given in [appendix](#). Note that **your solution must not copy the lists to vectors**. Instead, your solution must implement the algorithm directly with the lists.

- Overloaded operator `*=` such that `R *= S`; should be equivalent to $R = R \cap S$, i.e. the *intersection* of R and S is assigned to set R . Recall that the intersection $R \cap S$ is the set of elements in both R and S . For instance, if $R = \{1, 3, 4\}$ and $S = \{1, 2, 4\}$ then $R \cap S = \{1, 4\}$.
- Overloaded operator `--` such that `R -= S`; should be equivalent to $R = R - S$, i.e. the *set difference* of R and S is assigned to set R . Recall that the set difference $R - S$ is the set of elements that belong to R but do not belong to S . For instance, if $R = \{1, 3, 4\}$ and $S = \{1, 2, 4\}$ then $R - S = \{3\}$ ³.

The file `lab2.cpp` contains a test program for class `Set`. Feel free to add other tests to this file, though the original file must be used when presenting the lab.

² $S1 \leq S2$ tests whether set $S1$ is a subset of set $S2$, i.e. all elements of $S1$ also belong to $S2$.

³ The set difference $R - S$ is also known in the literature as the "*relative complement of S in R* ".

Code requirements

- All Set functions must have a linear time complexity, in the worst-case. You should also strive to implement set member functions by just traversing each list at most once.
- The comparison operators should iterate through each set no more than once.
- Do not spread the use of `new` and `delete` all over the code. Instead, define two private member functions which call `new` and `delete`.

```
// Insert a new Node storing v after the Node pointed by p
void insert_node(Node* p, int v);
```

```
// Remove the Node pointed by p
void remove_node(Node* p);
```

Class Set member functions should call the private member functions above, whenever possible, to add or remove a node⁴.

- The public class Set interface (given in `set.h`) cannot be modified. Other member functions can be added to class Set as private member functions, though.
- Finally, STL containers cannot be used to implement the member functions⁵. Therefore, solutions that involve copying lists into containers such as vectors are not accepted.

Exercise 2: time complexity analysis

This exercise consists in analysing the time complexity for each of the following statements. Use Big-Oh notation and write a **clear motivation** for your analysis. Assume that `S1` and `S2` are two Set variables and that `k` is an `int` variable. Assume also that `S1` has $n_1 > 0$ elements and that `S2` has $n_2 > 0$ elements.

- `S1 = S2;`
- `S1 * S2`
- `k + S1`

Make sure that the mathematical functions used to express time complexity clearly relate to variables n_1 and n_2 .

Presenting lab and deadlines

The exercises in this lab are compulsory and you should demonstrate your solutions during the lab session *Lab2 RE*. Read the instructions given in the [labs webpage](#) and consult the course schedule.

Necessary requirements for approving your lab are given below.

⁴ Possible exception can be when creating or destroying a list's dummy nodes in the constructors and destructor, respectively.

⁵ An exception applies to the Set class constructor, which creates a set using elements from an input `std::vector`.

- The code should be readable, properly indented, and follow good programming practices. Avoid overly complex functions and repeated code, as they reduce readability and increase the risk of bugs.
- Compiler warnings that may affect badly the program execution are not accepted, though the code may pass the given tests.
- There are no memory leaks neither other memory related bugs. See the appendix for recommended [memory monitoring tools](#).
- Your code must satisfy the given [code requirements](#).
- Bring your written answer for [exercise 2](#) for *Lab 2 RE* and be prepared to discuss it with the lab assistant. Unreadable answers will be rejected.

If your code for lab 2 has not been approved in the scheduled lab session *Lab2 RE* then it is considered a late lab. Late labs can be presented provided there is time in a another RE lab session. All groups have the possibility to present one late lab on the extra RE lab session scheduled in the end of the course.

Expected output

```
TEST PHASE 0: default and conversion constructor
TEST PHASE 1: constructor from a vector
TEST PHASE 2: copy constructor
TEST PHASE 3: operator=
TEST PHASE 4: is_member
TEST PHASE 5: cardinality and is_empty
TEST PHASE 6: equality and <=>
TEST PHASE 7: operator+=, operator*=:, operator-=
TEST PHASE 8: union, intersection, and difference
TEST PHASE 9: mixed-mode arithmetic
Success!!
```

Appendix

Algorithm: Merge two sorted sequences of unique elements

Goal: Merge two increasingly sorted sequences S1 and S2 into an increasingly sorted sequence R. None of the sequences S1, S2, or R have duplicates.

Example:

- S1 = [1, 3, 5, 8]
- S2 = [2, 3, 6, 8, 10]
- R = [1, 2, 3, 5, 6, 8, 10] (merged sequence, each value appears once)

Idea: Walk through both sequences once, always taking the smaller current item. When items are equal, take one and advance both sides.

1. Initialize

- Position at the first item of S1 (call it x if it exists).
- Position at the first item of S2 (call it y if it exists).

2. Merge while both sequences have items

- Compare x and y:
 - If $x < y$:
 - Insert x into R.
 - Advance S1 to its next item and update x.
 - If $y < x$:
 - Insert y into R.
 - Advance S2 to its next item and update y.
 - If $x == y$:
 - Insert x (or y) once into R.
 - Advance **both** S1 and S2 to their next items; update x and y.

3. Drain the remainder

- If S1 still has items, insert them in order into R.
- If S2 still has items, insert them in order into R.

4. Finish

- R now contains all unique items from S1 and S2 in increasing order.

Comparison operators in C++20

C++20 has introduced new rules concerning overloading comparison operators such as `operator==`, `operator!=`, `operator<`, `operator<=`, etc. With these new rules, for most cases, it's only required to overload two comparison operators, `operator==` and `operator<=>`, so that a class supports all six comparison operators. In addition, extra overloads are not required to support mixed-mode comparison, either. Thus, a lot of the boilerplate code that comparison operators introduced until C++20 has now disappeared.

In C++20, if an `operator==` is available for a class then `a != b` is automatically evaluated as `!a.operator==(b)`. In addition, comparison operator calls can be reversed, automatically. For example, `9 == a` evaluates to `a.operator==(9)` (thus, taking advantage of possible automatic conversion from `int` to the type of object `a`). Thus, C++20 understands that equality is symmetric.

A new operator, `operator<=>` (named three-way comparison operator or operator spaceship) has been added in C++20. This operator should return instances of one of the three comparison category classes defined in the library `compare`. These categories are: `std::strong_ordering` (used when all pairs of instances are comparable), `std::weak_ordering` (used when equality only defines an equivalence class), and `std::partial_ordering` (used when a pair of instances `a` and `b` can be incomparable). For instance, `operator<=>` should return instances of `std::strong_ordering` for ints, while `operator<=>` should return instances of `std::partial_ordering` for the class `Set` in this exercise. Consider a set `S1 = {1, 2, 8}` and a set `S2 = {3, 9}`. Then, neither `S1 < S2` (`S1` is not contained in `S2`) nor `S2 < S1` (`S2` is not contained in `S1`), i.e. `S1` and `S2` are simply not comparable.

More concretely, for class `Set`, the three-way comparison `S1<=>S2` should return:

- `std::partial_ordering::equivalent`, if `S1` and `S2` represent the same set;
- `std::partial_ordering::less`, if `S1 < S2`, i.e. all elements of set `S1` also belong to set `S2` but `S1` and `S2` are not the same set;
- `std::partial_ordering::greater`, if `S2 < S1`;
- `std::partial_ordering::unordered`, otherwise (i.e. sets `S1` and `S2` are incomparable sets).

If the three-way comparison operator is defined for a class, then it automatically gives support for all four comparison operators, `<`, `<=`, `>`, `>=` (i.e. no need to write code for these operators). For instance, `20 < S1` also compiles (assuming there is a conversion from `int` to `Set`) meaning that `20` is an element of set `S1`, i.e. `S1 > {20}`.

This section provides a brief introduction to comparison operators in C++20. More detailed explanations can be found on many websites, in C++20 books, and AI chatbots can also generate short tutorials on the topic.

Memory monitoring tools

C++ programs can be affected by bugs that corrupt memory such as [buffer overflows](#), accesses to a [dangling pointer](#) (use-after-free), or memory leaks. It is important to detect this type of bug and eliminate them, though it may not be a trivial task.

Specific memory monitoring software tools can help the programmers to find out if their programs suffer from memory corruption bugs. There are several tools which you can install and try for free. Note that no tool will detect all memory bugs in the code (i.e. all tools have limitations). Thus, using several tools and inspecting carefully the code is the best strategy. Some of these tools are listed below. Take the opportunity and try some of them.

- [AddressSanitizer](#) (ASan) is a tool to detect memory related problems in the code, though it cannot detect memory leaks on Windows. ASan is supported by [Microsoft C/C++ compiler \(MSVC\)](#). If you have created the Visual Studio project/solution using the CMake file provided with this lab then the address sanitizer is already enabled. Otherwise, you need to enable it manually.
- [Valgrind](#) available for Linux.